



Data as a New Mechanism for Strengthening Accumulation and Economic Control

The literature on the concept of 'informational capitalism' has helped highlight the growing role of information and digital technologies in reshaping the global economy. Yet most of these approaches whether Manuel Castells's theorization of the 'network society', the theorists of cognitive capitalism, or even the more recent literature on 'surveillance capitalism' in Shoshana Zuboff's work, or 'platform capitalism' in Nick Srnicek's have largely remained confined to sociological, technological or behavioral description, focusing on networks, platforms and techniques of behavioral extraction, without systematically situating these transformations within the dynamics of global political economy and the power relations that govern the evolution of the capitalist system in its monopoly phase.

From a critical political-economy perspective, and in line with the analysis that explains the monopoly phase of capitalism through the transformation of industrial capital into dominant monopoly finance capital governing the world economy via capital export and the division of the world among major monopolies, digital technology cannot be regarded as an autonomous force that makes history, but rather as part of the system of productive forces that develops within specific economic, social and political relations, and as an expression of the current phase of capital concentration. Technology does not operate independently of the logic of capital accumulation; it contributes to reshaping that logic and expanding its scope, while also redistributing economic and cognitive power among actors at the global level.

Just as the steam engine did not create industrial capitalism but served as a tool for accelerating production and deepening capital accumulation, and just as the banking export of capital did not give rise to imperial domination but was its operational instrument, artificial intelligence, big data, cloud computing and digital platforms did not create informational capitalism they have become new tools for reorganizing production, managing markets, strengthening control over material and immaterial resources, and expanding the scope of capitalist accumulation across national borders.

It therefore appears necessary to move from the concept of 'informational capitalism' which describes a general technical condition to a concept that more accurately captures the nature of the current phase and the power relations that structure it: 'monopoly digital capitalism'. The defining feature of this phase is not merely the widespread diffusion of information or digital technologies, but the unprecedented concentration of ownership of digital infrastructure in the hands of a small number of transnational firms that dominate cloud servers, search engines, algorithms, artificial intelligence, e-commerce platforms, telecommunications networks and global databases a concentration that reproduces, with new tools, the very monopolistic structure that marked the finance-capital phase at the turn of the twentieth century.

This concentration reflects more than a neutral technical development: it expresses the continuation, in new forms, of the logic of capital concentration, with control over data, digital knowledge and technological infrastructure having become one of the principal sources of economic and political influence in the digital age rivalling, and at times surpassing, the influence historically associated with control over raw materials and financial markets.



From the Monopolization of the Means of Production to the Monopolization of Digital Infrastructure

Political-economy analysis shows that the development of capitalism has historically been accompanied by a concentration of ownership of the means of production and an intensification of economic monopolies, culminating in the merger of industrial capital with banking capital into a dominant finance capital. In the digital era, this phenomenon has expanded to encompass what may be termed the 'digital infrastructure of production', which is no longer limited to material means of production but now also includes the tools for producing knowledge, managing data and making automated decisions.

Major digital firms no longer merely hold capital or control specific economic sectors; they have come to dominate new strategic elements: digital platforms, algorithms, cloud infrastructure, search engines, artificial-intelligence systems and databases. Researcher Jathan Sadowski describes these elements as having a dual character: data as capital, and digital infrastructure as fixed capital comparable to the factory or the machine in the industrial economy. These digital infrastructures now play an increasingly decisive role in organizing information flows, steering markets, and setting the terms of access to knowledge and resources, thereby shaping the trajectories of production, consumption and decision-making.

This shift reveals the movement of economic monopoly into a new domain: control over digital infrastructure, which has become one of the fundamental pillars of the contemporary economy. Owning this infrastructure confers not only a technical capacity to manage data and information, but also growing economic and political power, given the central role that digital platforms now play in organizing economic and social relations including in strategic sectors such as agriculture and food, where supply-chain management platforms and digital traceability systems are becoming new instruments for reconfiguring sovereignty over productive resources.

Monopoly is thus no longer tied solely to ownership of material means of production; it now also encompasses the capacity to control the digital means that produce knowledge, gather data and organize information flows. This reflects a profound shift in the nature of power relations within the global economy, with control over digital infrastructure having become one of the principal sources of influence in the current period one of the contemporary forms of the logic of 'dividing up the world' among major monopolies, except that the object of this division today is no longer only markets and colonies, but digital infrastructure and the informational space itself.

This form of monopoly differs from traditional industrial monopolies in that it is not confined to ownership of the means of production but extends to control over the digital infrastructure on which the global economy depends making information, data and algorithms strategic resources of an importance comparable to energy, capital or natural resources.

Data as a Strategic Means of Production: Raw Material or Fixed Capital?

In the industrial economy, production rested essentially on three principal elements: land, labour and capital. In the digital economy, data has come to fulfil an increasingly strategic function in organizing production, steering markets and shaping decision-making.

Critic literature nevertheless differs over the precise position of data within the production process. While some researchers, such as Nick Couldry and Ulises Mejias in their analysis of 'data colonialism', tend to regard data as a raw material 'extracted' from everyday social life in ways comparable to how historical colonies



were exploited as sources of raw materials, Sadowski argues that data has a dual quality: on the one hand, it is a raw material subject to extraction; on the other, once processed and transformed into predictive models and operational algorithms, it becomes itself a form of fixed capital embedded in the production process.

From a critical political-economy perspective, a distinction must be drawn between data as an economic resource and labour as the primary source of value. Data does not produce economic value autonomously; it acquires importance only through its deployment in improving production, forecasting economic behavior, guiding investment and enhancing profit-making capacity. Data has thus become an essential condition for expanding the processes of capital accumulation, which justifies regarding it as a component of the digital means of production without, however, displacing human labour as the sole source of value.

Digital Means of Production

These include:

- Big data
- Algorithms
- Artificial-intelligence applications
- Cloud computing
- Telecommunications networks
- Satellites
- Digital platforms
- Data infrastructure

Control over these elements has become one of the principal determinants of economic power in the digital age, as it grants those who hold it the capacity to organize production, steer markets, control information flows and reshape global value chains.

From Surplus Value to Data Surplus

Political-economy analysis shows that capital accumulation rests fundamentally on the extraction of surplus value generated by human labour. The digital economy, however, has added a new mechanism to reinforce accumulation: the transformation of data produced by individuals and communities into a new source of profit and control a mechanism that recalls, in renewed forms, Dallas Smythe's thesis on the 'audience commodity', as well as Christian Fuchs's later analysis of the unpaid 'digital labour' produced by platform users, who are often unaware that they are thereby participating in a process of value creation.

Digital platforms do not merely organize economic and social exchanges: they collect vast quantities of data on user behavior, consumption patterns, social relationships, geographic location, individual preferences and local knowledge, which they then analyse and redeploy within new economic models such as targeted advertising, predictive analytics and digital services. Data has thus become a strategic resource that contributes to expanding the scope of accumulation and strengthens capital's capacity to anticipate economic and social behavior and control markets.

This shift gives rise to a new form of value extraction within the digital economy: platform-owning firms benefit from the data that individuals and communities generate through their daily and productive activities without direct compensation, while the capacity to control this data and determine its economic value remains concentrated among the actors who own the digital infrastructure. This does not mean that data



has become an autonomous source of value displacing labour, but rather that it constitutes an additional element that reinforces accumulation processes and reshapes power relations between the producers of data: individuals, farmers, artisans, local communities and those who own the means to collect, process and monetize it.

This concept does not replace that of surplus value; it clarifies how data has become an additional instrument for expanding accumulation processes and strengthening economic domination under monopoly digital capitalism.

Digital Colonialism: A Contemporary Extension of International Relations of Domination

If the classical political-economy literature explaining the monopoly phase of capitalism through capital export and the division of spheres of influence among major powers accounted for the expansion of global capitalism through control over natural resources, markets and capital, current digital transformations reveal a new form of domination based on control over data, digital infrastructure, algorithms and global platforms.

Digital colonialism may accordingly be defined as the process of reproducing relations of dependency and domination through control over data, digital infrastructure, algorithms and technology platforms, whereby economic, cognitive and decision-making power is transferred from peripheral societies and states to the dominant centers of the global digital economy.

This form of domination extends the transformations undergone by the global capitalist system, in which mastery of digital knowledge and data has become no less important than historical control over land and natural resources, and technology itself has become a field for reproducing power relations and inequality at the international level. This shift can also be read in light of Samir Amin's theses on unequal exchange and the centrality of the global capitalist system: center countries occupy the position of producers of technology and digital standards, while peripheral countries including those of the Arab region and Africa — are relegated to the position of raw data exporters and dependent importers of pre-established digital services and standards, in a structural reproduction of the dependency relationship described by classical dependency theory, albeit through new tools and forms.

Digital colonialism thus does not abolish classical international relations of domination based on control over land, resources and markets; rather, it adds a further layer of dependency normative, technical and cognitive whereby transnational firms, through their ownership of digital infrastructure, determine the conditions of knowledge production and set the technical standards imposed on peripheral countries without their genuine participation in shaping them. This opens the way to questioning 'digital governance' policies presented internationally as neutral solutions, when in substance they serve as instruments for consolidating this new international division of digital labour.